Lifecycle Management

Use **VCF Operations** as a VI administrator to manage the lifecycle of the management and core components in VCF, including downloading binaries and updating VCF fleet and VCF Instances.

For VCF, you use VCF Operations for these capabilities. For vSphere Foundation, you perform the supported operations on the individual management components.

In This Chapter

- Binary Management for VMware Cloud Foundation
- Lifecycle Management of VCF Components
- Upgrading VMware Cloud Foundation Workload Domains to 9.1.x

Getting to Know VCF

To apply this documentation, you must be acquainted with the VCF Product Overview, VCF Design, and VCF Release Notes.

VCF Component Classification

VCF components are classified based on the level at which they operate. VCF Management components operate on the fleet level, while VCF core components operate on the instance level.

VCF Management Components	VCF Core Components
• VCF Operations – License server – Cloud proxy • VCF management services • VCF Automation • VCF Operations for networks	• SDDC Manager • vCenter • NSX Manager • ESX Optional: • VCF Operations HCX*

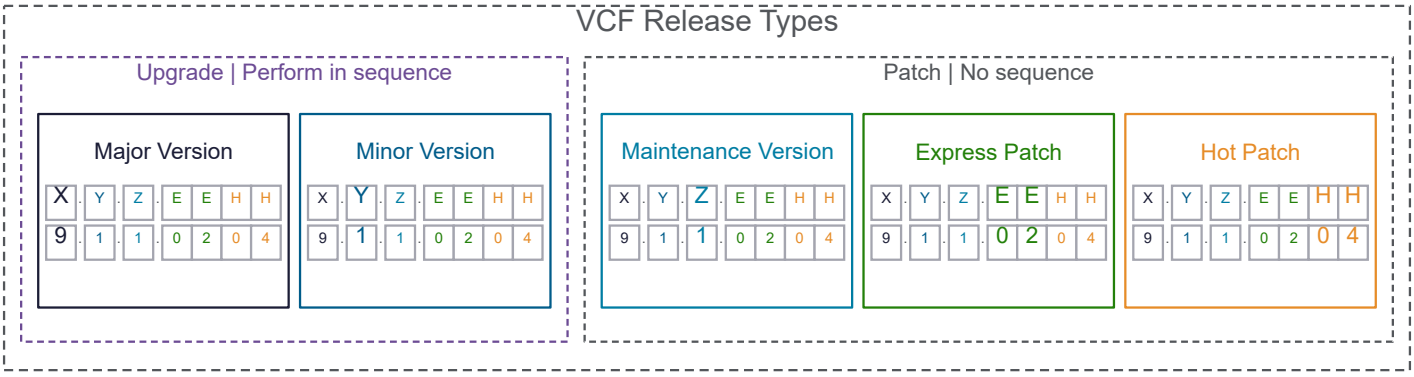
* VCF Operations HCX is an optional component. Lifecycle operations for VCF Operations HCX are performed with VCF core components.

VCF Release Types

The VCF release policy defines four distinct release types, each with a specific purpose, scope, and set of lifecycle rules.

Type	Description	Approximate Release Cadence
Major	A major release, such as 9.0.0.0 or 10.0.0.0, represents the most significant evolution of the VCF platform. It introduces major features and changes that can redefine the platform's capabilities. This is the only release type that can introduce breaking changes or remove existing functionality. However, this is governed by a strict deprecation policy designed to protect enterprise stability: a feature must be officially announced for deprecation in a prior major release before it can be removed in a subsequent one. Major releases are fully synchronized, meaning all VCF components are upgraded in lockstep as part of a single release. They follow a unified consumption pattern, requiring a full-stack upgrade.	3 years
Minor	A minor release, such as 9.1.0.0 or 9.2.0.0, is the primary mechanism for delivering new features and enhancements within a major version. These releases include features and are designed to be functionally additive. Minor releases aim to be non-disruptive from a compatibility and API perspective. Like major releases, minor releases are fully synchronized and follow the unified consumption pattern. The policy that "All components are upgraded in a major and minor release" applies here, ensuring the entire platform advances to a new, consistent feature level.	9 months
Maintenance	A maintenance release, such as 9.0.1.0 or 9.0.2.0, is a scheduled, cumulative upgrade, focused on enhancing platform stability and support. Maintenance releases provide bug and security fixes, hardware enablement changes, such as support for servers, CPUs, devices, driver updates, guest OS support, and backward-compatible new features. They are limited in scope to minimize the operational impact. You can pick and choose which components to apply based on the impact on your environment and operational costs. For example, if you run into a critical issue with vCenter, you do not need to apply a patch to ESX that is irrelevant to your environment. Maintenance releases are synchronized among VCF components with a new Bill Of Materials.	3 months
Express Patch	Express patch releases, such as 9.0.1.0100 or 9.0.1.0300, provide component-level, time-sensitive fixes to security and catastrophic issues, impacting business operations, that must be released between major, minor, and maintenance releases. Express patches are released at the component level only and are not synchronized across VCF components. Express patches do not require a new implementation of a Bill of Materials. Express patch releases for a given component do not always follow a strict sequence. Each patch release increments the second digit of the fourth version field (x.y.z.a) but can skip values. For example, the first patch might be 9.1.0.0100, and the next patch for that component might be 9.1.0.0300 or 9.1.0.0500 and so on.	As needed
Hot Patch	Hot patch releases, such as 9.0.1.0101 or 9.0.1.0112, provide on-demand, component-level fixes for issues with limited impact where standard express patch payloads are not viable. Hot patches are released at the component level only and are not synchronized across VCF components. Hot patches do not require a new implementation of a Bill of Materials. Hot patches are not cumulative, a higher version number does not include all previous content. Security fixes are prohibited in hot patches unless the fixes were already released in a generally available patch.	As needed

VCF release types diagram showing five release component types in table format, grouped under Upgrade and Patch categories within VCF release types.



VCF Lifecycle Operation Types

VCF defines two types of lifecycle operations to increase your VCF version:

Lifecycle Operation Type	Description	Release Type	Version Change
Upgrade	An upgrade is defined as a change to the major or minor version of a VCF component. Upgrades are synchronized across all VCF components and require a new Bill of Materials (BoM), meaning the entire platform advances to a new, consistent feature level in lockstep.	Major	9.0.0.0000 to 10.0.0.000
		Minor	9.0.0.0000 to 9.1.0.0000
Patch	A patch is defined as a change to the maintenance or patch version of a VCF component, while the major and minor versions remain unchanged. There is no enforced patch order for maintenance or patch releases. Any VCF component can be patched independently of other components. However, specific component dependencies may apply for certain releases, requiring one component to be patched before another, as documented in the release notes.	Maintenance	9.0.0.0000 to 9.0.1.0000
		Express Patch	9.0.1.0000 to 9.0.1.0200
		Hot Patch	9.0.1.0000 to 9.0.1.0003

Lifecycle Management in VCF Operations

VCF Operations provides a comprehensive set of tools for managing the lifecycle of VCF components. You can use VCF Operations to download and manage upgrade, patch, and install binaries. You can also manage ESX components and vSphere Lifecycle Manager images and deploy, upgrade, and patch additional VCF components.

For VCF 9.0 and later, VCF Operations manages major, minor, maintenance, and patch releases for your VCF environment. With the introduction of VCF management services in VCF 9.1, VCF Operations now uses the fleet lifecycle, SDDC lifecycle, and software depot components to orchestrate lifecycle operations on both fleet and instance-level components. In addition, the fleet lifecycle and SDDC lifecycle components now replace the VCF Operations fleet management appliance.

With VCF management services, VCF Operations now uses a declarative lifecycle model. You define a target version for your VCF environment and VCF Operations orchestrates the upgrade or patch processes, including planning, required binaries, and additional dependencies. This approach consolidates management resources and ensures high availability for lifecycle operations.

SDKs, APIs, and CLI for VCF Administration

You can build, operate, and manage your VCF private cloud by using the VCF SDK and APIs, and VCF PowerCLI. See Administration SDKs, APIs, and CLI.

Binary Management for VMware Cloud Foundation

With VCF 9.1, you perform lifecycle management of VCF components by using the VCF Operations UI and the software depot component, which handles binaries for all VCF components.

Software Depot Overview

A software depot is VCF 9.1 or later component that serves an internal store for installation, upgrade, patch binaries, and Open Container Initiative (OCI) images. A software depot can operate on the fleet-level or optionally on an instance-level. Each VCF Instance can have one software depot instance that is assigned to it.

First Software Depot (Fleet-Level)

Each VCF fleet has a fleet-level software depot that is deployed on the first VCF Instance and can provide binaries for all VCF Instances in the fleet. When deploying a new VCF Instance version 9.1 or later, the fleet-level software depot is automatically assigned to the instance. When deploying a new VCF fleet or upgrading an existing one to version 9.1 or later, the depot settings from VCF Installer or SDDC Manager are transferred to the fleet-level software depot.

Secondary Software Depot (Instance-Level)

If the network latency between the software depot of your first VCF Instance and another VCF Instance exceeds 150 ms, you can deploy a secondary local software depot within the other VCF Instance. See Deploy a Secondary Software Depot Instance. After deploying a secondary software depot, assign it to a VCF Instance. See Assign a Software Depot to VCF Instances.

Offline Depot

If your VCF environment does not have access to the Internet, you can configure an internal web server to host binaries, and ESX component data. Starting with VCF 9.1.1, you can host also artifacts. By using the VCF Download Tool, the web server connects to the Broadcom online repository, downloads installation, upgrade, and patch binaries, and hosts them for use in your environment.

VCF Download Tool

The VCF Download Tool is a command-line utility that is used to download and manage binaries and metadata for VMware Cloud Foundation platforms in environments without Internet access. The tool provides commands for downloading, uploading, listing, and cleaning up VCF component and ESX binaries, supporting the management of VCF components required for VMware Cloud Foundation releases.

Configure a Software Depot Connection Mode

You can configure each software depot instance in one of the three available modes to access binaries, hardware compatibility data, software interoperability data, and lifecycle metadata. The mode depends on how you access the Internet in your environment.

- If you plan to use software depot in connected mode, but your environment does not have direct access to the Internet, verify you configured a proxy server. See Configure a Proxy Server for VCF Management Services Components and VCF Automation.

- Verify that your VCF Business Services console role has sufficient permissions. See Access Management.

Table 1087: Software Depot Modes

Connection Mode	Description
Connected	Use this option if your environment has access to the Internet, either directly or through a proxy server. Register your software depot with VCF Business Service console to directly download binaries, hardware compatibility data, software interoperability data, and lifecycle metadata.
Offline Depot	Use this option if your environment cannot connect to the Internet, but you have a privately owned server that you use as an offline depot. You use the VCF Download Tool to download the required binaries to your offline depot. In this mode, software depot does not download binaries locally. Instead, it proxies requests from VCF components to the privately owned server. For more information, see Download Binaries to an Offline Depot by Using the VCF Download Tool.
Disconnected	Use this option if your environment cannot connect to the Internet and you have no offline depot. You use the VCF Download Tool to download the required binaries to a computer with Internet access. You then manually upload the binaries to the software depot. For more information, see Download Binaries to Software Depot in Disconnected Mode by Using the VCF Download Tool.

For information about the URLs that must be accessible to download binaries, see Public URLs Required for Online Functionalities for VCF and vSphere Foundation.

To delete binaries from software depot in connected or offline depot mode, on the **Binary Management** tab of your VCF Instance, you can select the downloaded binaries and click **DELETE DOWNLOAD**. To delete binaries in disconnected mode, use the `cleanup` command. See VCF Download Tool Binaries Commands and Options.

If you deployed VCF or vSphere Foundation 9.1 or later, your online or offline depot configuration and authentication is transfered from VCF Installer to software depot and your software depot fleet-level instance is configured. If you upgraded from VCF 5.2 or 9.0.x to version 9.1 or later, follow this procedure to configure your software depot instance.

Note:

In environments where SSL/TLS interception is enabled, software depot configured in connected mode might not establish a trusted connection to the Broadcom authorization server. For more information on adding your proxy's CA certificate to the truststore of software depot, see Broadcom knowledge base article 442978.

Configure software depot.

Connected

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
2. Select **Build** > **Software Depot**.
3. To configure software depot, click **Edit**.
4. Select **Connected** and click **Next**.
5. Obtain and enter an activation code.
   a. Copy the software depot ID.
   b. Log in to the VCF Business Services console.
   c. Register the component with the correct tenant.
      a. In the upper-left corner of the console, next to the tenant name, click the arrow to expand the section.
      b. From the drop-down menu, select the correct tenant name and siteID that map to your VCF or vSphere Foundation entitlement.
   d. Navigate to **Software Depot Registrations** > **New Registration** and use your software depot ID to obtain an activation code for your depot.

e. Enter the activation code that you obtained from the VCF Business Services console and click **Configure**.

6. Click **Finish**.

Offline Depot

1. If your offline depot web server is connected over HTTPS, you must import your TLS certificate to the certificate store of SDDC Manager.
   a. SSH to SDDC Manager.
   b. Log in with `vcf` user.
   c. Change to root with `su`.
   d. Download the certificate from your offline depot web server and copy the certificate to the SDDC Manager appliance.

      ```
      openssl s_client -connect depot_server_FQDN:443 < /dev/null | sed -ne '/-BEGIN
       CERTIFICATE-/,/-END CERTIFICATE-/p' > /tmp/server.crt
      ```

   e. Import your certificate into the certificate store.

      ```
      keytool -import -trustcacerts -file /tmp/server.crt -keystore /etc/
      alternatives/jre/lib/security/cacerts -alias offline_depot_server -storepass
       changeit -noprompt
      ```

   f. Restart the lifecycle manager service.

      ```
      systemctl restart lcm
      ```

2. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
   a. Select **Build** > **Software Depot**.
   b. To configure software depot, click **Edit**.
   c. Select **Offline Depot** and click **Next**.
   d. Enter the URL for the offline depot.
      If your offline depot requires authentication, toggle the **Authentication** setting to the on position and enter the user name and password.

      Note:

      VMware Cloud Foundation does not support proxy servers with Kerberos or NTLMv2 authentication.

   e. Click **Finish**.

Disconnected

1. Select **Disconnected** and click **Next**.
2. Click **Finish**.

After software depot connects successfully, you can download binaries. See Download Binaries to Software Depot.

- To view available binaries, click **Build** > **Lifecycle**, select a VCF Instance, and click **Binary Management** > **Upgrade Binaries**.
- To download the required binaries, see Download Binaries to Software Depot. For example, to complete an upgrade process, you must download the vCenter, NSX, and ESX, and other required VCF component binaries.

Download Binaries to Software Depot

After you configure software depot, you can download installation, upgrade, and patch binaries.

- Verify that software depot is configured. For more information, see Configure a Software Depot Connection Mode.

When you download binaries, software depot verifies that the file size and checksum of the downloaded binaries match the expected values.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the Administrator role.
2. Select **Build** > **Lifecycle**.
3. Select a VCF Instance and click **Binary Management**.
4. From the **VCF Version** drop-down, select a target version.
5. Click the **Upgrade Binaries**, **Patch Binaries**, or **Install Binaries** tab to view available binaries.
   - To manually upgrade the binaries list, click **Sync Now**.

- To view more information about a binary, click the component name.

6. To download ESX component data, under **ESX Components**, click **Synchronize Now**.

   The downloaded content includes the Broadcom published base images, add-ons, VMtools, and so on. To download or import third-party ESX binaries, see Download or Import Third-Party ESX Binaries to Software Depot for Connected Mode.

7. From the list of available components, select one or more components and click **Download**.

   Alternatively, you can click **Schedule Download** and select another date and time for the binary download.

Download or Import Third-Party ESX Binaries to Software Depot for Connected Mode

In VCF 9.1 and later, software depot acts as a repository for vSphere Lifecycle Manager and it is automatically added as a download source for ESX component data in vSphere. If you use software depot in connected mode and you require ESX binaries from a third-party vendor, you must perform additional steps.

vSphere Lifecycle Manager must have all the required ESX component data before you can create an image to use to upgrade ESX. To add third-party binaries, you can configure UMDS to connect to the websites of third-party vendors or import third-party offline zip files.

1. By using the `vcf` user account, SSH in to the SDDC Manager appliance.
2. Navigate to `/opt/vmware/vmware-umds/bin`.
3. Add third-party ESX binaries URL or offline zip bundle.

Option	Description

Configure UMDS to connect to the websites of third-party vendors to download ESX component data.	1. Add the third-party vendor URL.
	`vmware-umds -S --add-url https://host_url/index.xml --url-type HOST`
	2. Download ESX component data.
	`./vmware-umds -D`
	Replace *host_url* with the third-party vendor URL.

Import a single third-party add-on or component as an offline zip bundle.	`vmware-umds -I --offline-bundle path_to_zip_file` Replace *path_to_zip_file* with the absolute path to the zip bundle.

Set Up an Offline Depot

To use VCF Installer or software depot with an offline depot, you must configure an internal web server to host VCF binaries and ESX component data. You use the VCF Download Tool to connect to the Broadcom depot and download the binaries to the internal web server.

The offline depot web server must meet the following requirements:

- Operating System: A Linux or Windows virtual machine. For Linux, any distribution is supported (for example, CentOS, RHEL, Ubuntu). For example, you can use Photon OS. See https://packages.broadcom.com/photon/.
- Storage: A dedicated disk for storing offline depot content. Minimum recommended size: 1 TB.
- Web Server and SSL: A web server to serve the offline depot content over HTTP/HTTPS. For HTTPS, the server must support TLSv1.2 and TLSv1.3. The choice of web server is flexible. For example: Apache HTTP Server, NGINX, and so on.
- Networking and Access: The virtual machine must have a static IP address. DNS records for the offline depot server are recommended but not required.
- User Access: A dedicated non-root user on the virtual machine. The user must have read/write access to the offline depot directory and appropriate permissions to run VCF Download Tool commands.

The following procedure describes how to set up an Apache HTTP Server on a Photon OS 5 virtual machine. Depending on your environment, you might need to modify some of the commands. For example, if corporate policies require you to use a different operating system or web server.

The procedure uses `/var/www/html` as the document root directory for the web server. The directory must exist or you must manually create it by running the `mkdir -p /var/www/html` command. Alternatively, to use a different document root directory, in the procedure, replace `/var/www/html` with your preferred root directory.

The following procedure uses *`depot`* as an example name of the depot VM and *`depot.rainpole.io`* as the offline depot URL.

For a guided step-by-step video demonstration, see VCF 9.0 Demo Offline Depot.

With VCF 9.1 and later, offline depot is supported with the following HTTP and HTTPS protocol authentication.

- HTTPS with basic authentication
- HTTPS without authentication
- HTTP without authentication

1. Deploy the Photon OS OVA file on a VM.

   Do not power on the VM.

2. Add a hard drive with a capacity of 1 TB or more.
3. Power on and log in to the Photon OS 5 VM.
4. Specify networking details.
   a) Navigate to `/etc/systemd/network/` and create a `.network` file.

      For example, *`10-static-en`*`.network`.
   b) In a text editor, open the `.network` file and add your network configuration.

      For example:

      ```
      [Match]
      Name=eth0

      [Network]
      Address=172.16.10.14/24
      Gateway=172.16.10.1
      DNS=172.16.10.4
      DNS=172.16.10.5
      ```

   c) Set the correct permissions for the `.network` file.

      ```
      chmod 644 10-static-en.network
      ```

5. Update the hostname by using the `hostnamectl set-hostname` *`offlinedepotURL`* command, where is *`offlinedepotURL`* is your offline depot URL.

   For example:

   ```
   hostnamectl set-hostname depot.rainpole.io
   ```

6. Restart the network and DNS lookup services.

   ```
   systemctl restart systemd-networkd
   systemctl restart systemd-resolved
   ```

7. Install Apache HTTP Server, update the operating system, and reboot the VM.

   ```
   tdnf install httpd tar jq --assumeyes
   tdnf update --assumeyes
   reboot
   ```

8. Format the disk you added to the VM.

   ```
   mkdir -p /var/www/html
   mkfs.ext4 /dev/sdb
   echo "/dev/sdb /var/www/html ext4 defaults 0 2" >> /etc/fstab
   mount -a
   ```

9. Prepare the SSL Certificate.

   By following the commands, you create a directory and the private key.

   ```
   mkdir /root/http-certificates
   openssl genpkey -out /root/http-certificates/server.key -algorithm RSA -pkeyopt rsa_keygen_bits:2048
   ```

   a) Provide SAN information, create a configuration file with the required attributes, for example, `conf.cfg`, and add the file to the `openssl req` command.

      Replace the following example values in the template with your own values: *`conf.cfg`*, *`SE`*, *`Stockholm`*, *`Stockholm`*, *`Rainpole`*, *`IT`*, *`depot.rainpole.io`*, and *`172.16.10.11`*.

      Note:

      OpenSSL does not prompt you for Subject Alternative Names (SANs) during the certificate request (CSR) creation process. SANs are treated as extensions, rather than standard subject information fields.

      ```
      cat << 'EOF' > /root/http-certificates/conf.cfg
      [req]
      distinguished_name = req_distinguished_name
      req_extensions = req_ext
      prompt = no

      [req_distinguished_name]
      C = SE
      ST = Stockholm
      L = Stockholm
      O = Rainpole
      OU = IT
      CN = depot.rainpole.io

      [req_ext]
      subjectAltName = @alt_names

      [alt_names]
      IP.1 = 172.16.10.11
      DNS.1 = depot.rainpole.io
      EOF
      ```

   b) Create a `.csr` file by using the `.key` and `.cfg` files.

      ```
      openssl req -new -key /root/http-certificates/server.key -out /root/http-certificates/request.csr -
      config /root/http-certificates/conf.cfg
      ```

10. Sign the certificate.

Signing Method	Steps

External Certificate Authority	1. Sign `request.csr` using your organization's external certificate authority.
	2. Combine the machine, intermediate, and root certificates into one `server.crt` file (PEM format), in this order:
	`-----BEGIN CERTIFICATE-----`
	`[Machine Cert]`
	`[Intermediate Cert]`
	`[Root Cert]`
	`-----END CERTIFICATE-----`
	3. Copy `server.crt` to `/root/http-certificates/`.

VMware Certificate Authority (VMCA)	1. Transfer `request.csr` to the vCenter (for example, to `/root/`).
	2. SSH into vCenter as `root` and run the following commands:
	`openssl x509 -req -days 365 -in /root/request.csr \`
	`-CA /var/lib/vmware/vmca/root.cer \`
	`-CAkey /var/lib/vmware/vmca/privatekey.pem \`
	`-CAcreateserial -out /root/server.crt -sha256 \`
	`-extensions req_ext -extfile conf.cfg`
	`cat /var/lib/vmware/vmca/root.cer >> /root/server.crt`
	3. Copy `server.crt` to `/root/http-certificates/`.
	4. Clean the `.crt` certificate files.
	`rm -f /root/server.crt /root/request.csr`

11. Move the signed `.key` and `.crt` certificate files to the Apache configuration directory.

```
mv server.key server.crt /etc/httpd/conf/
chmod 0400 /etc/httpd/conf/server.key /etc/httpd/conf/server.crt
chown root:root /etc/httpd/conf/server.key /etc/httpd/conf/server.crt
```

12. To configure the Apache server, edit the `/etc/httpd/conf/httpd.conf` and `/etc/httpd/conf/extra/httpd-ssl.conf` files.
    a) Run the following commands to enable SSL modules and replace the `operations@rainpole.io` and `depot.rainpole.io:443` example values with your own.

```
sed -i 's|#LoadModule ssl_module|LoadModule ssl_module|' /etc/httpd/conf/httpd.conf
sed -i 's|#LoadModule socache_shmcb_module|LoadModule socache_shmcb_module|' /etc/httpd/conf/httpd.conf
sed -i 's|#Include conf/extra/httpd-ssl.conf|Include conf/extra/httpd-ssl.conf|' /etc/httpd/conf/httpd.conf
sed -i 's|DocumentRoot "/etc/httpd/html"|DocumentRoot "/var/www/html"|' /etc/httpd/conf/httpd.conf
sed -i 's|<Directory "/etc/httpd/html"|<Directory "/var/www/html"|' /etc/httpd/conf/httpd.conf
sed -i 's|DocumentRoot "/etc/httpd/html"|DocumentRoot "/var/www/html"|' /etc/httpd/conf/extra/httpd-ssl.conf
sed -i 's|ServerAdmin you@example.com|ServerAdmin operations@rainpole.io|' /etc/httpd/conf/extra/httpd-ssl.conf
sed -i 's|ServerName www.example.com:443|ServerName depot.rainpole.io:443|' /etc/httpd/conf/extra/httpd-ssl.conf
```

b) In the `/etc/httpd/conf/httpd.conf` file, locate the `DocumentRoot "/var/www/html"` section and change `Required all denied` to `Required all granted`.

c) In the `/etc/httpd/conf/extra/httpd-ssl.conf` file, locate the `</VirtualHost>` tag and above the tag, paste the following configuration code.

```
<Directory /var/www/html/PROD/COMP>
        AuthType Basic
        AuthName "Basic Authentication"
        AuthUserFile /etc/httpd/conf/.htpasswd
        require valid-user
</Directory>
<Directory /var/www/html/PROD/metadata>
        AuthType Basic
    AuthName "Basic Authentication"
    AuthUserFile /etc/httpd/conf/.htpasswd
    require valid-user
</Directory>
<Directory "/var/www/html/PROD/COMP/Compatibility/VxrailCompatibilityData.json">
        # VxRail VVS Cookie Validation (VCF 5.0)
        <If "%{HTTP:Cookie} == 'ngssosession=ngsso-token' ">
        Require all granted
        </If>
</Directory>
<Directory /var/www/html/PROD/vsan/hcl>
        Require all granted
</Directory>
        # Those Alias statements are needed only for VCF 5.1.0.0.
        Alias /products/v1/bundles/lastupdatedtime /var/www/html/PROD/vsan/hcl/lastupdatedtime.json
        Alias /products/v1/bundles/all /var/www/html/PROD/vsan/hcl/all.json
        # Needed only if UMDS downloads are presented
<Directory /var/www/html/umds-patch-store>
        Require all granted
</Directory>
```

13. Configure the authentication.

    Replace *`Username`* with the user name that VCF Operations and SDDC Manager use to authenticate against the depot.

```
htpasswd -c /etc/httpd/conf/.htpasswd Username
chown apache /etc/httpd/conf/.htpasswd
chmod 0400 /etc/httpd/conf/.htpasswd
```

14. Test the configuration, enable, and start the Apache HTTP server.

```
httpd -t
systemctl start httpd
systemctl status httpd
systemctl enable httpd
```

15. To open the HTTPS port in the firewall, edit `/etc/systemd/scripts/ip4save` file.

    In the file, above `COMMIT`, add the following text.

```
            -A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
```

    For example:

```
# init
```

```
*filter
:INPUT DROP [0:0]
:FORWARD DROP [0:0]
:OUTPUT DROP [0:0]
# Allow local-only connections
-A INPUT -i lo -j ACCEPT
-A INPUT -m conntrack --ctstate RELATED,ESTABLISHED -j ACCEPT
#keep commented till upgrade issues are sorted
#-A INPUT -j LOG --log-prefix "FIREWALL:INPUT "
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A OUTPUT -j ACCEPT
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
COMMIT
```

16. Restart the service.

```
systemctl restart iptables
```

17. Open a web browser and verify access to `https://<offline-depot-fqdn>/`.
18. Clean up the temporary certificate generation files.

```
rm -rf /root/http-certificates
```

19. Remove the default Apache index file.

```
rm -f /var/www/html/index.html
```

20. Set ownership and permissions for the document root directory.

```
chown apache -R /var/www/html/
find /var/www/html -type d -exec chmod 0500 {} \;
find /var/www/html -type f -exec chmod 0400 {} \;
chmod 755 /var/www/
chmod 755 /var/www/html/
```

- For VCF Installer, connect to the offline depot and download installation binaries. See Connect VCF Installer to Broadcom or an Offline Depot and Download Binaries and Download Installation Binaries to an Offline Depot.
- For software depot, configure an offline depot connection mode and download binaries. See Configure a Software Depot Connection Mode and Download Binaries to an Offline Depot by Using the VCF Download Tool.

Download Binaries to an Offline Depot by Using the VCF Download Tool

You can use the VCF Download Tool to download VCF component binaries and ESX component data to an offline depot and connect software depot to it.

- Verify that you have configured a web server that serves as an offline depot. See Set Up an Offline Depot.
- Verify that the machine where you plan to download and use VCF Download Tool has Internet connectivity (either directly or through a proxy) to download the binaries and other required files.
- Verify that the offline depot has TCP keepalive configured in your SSH client to prevent socket connection timeouts when using the VCF Download Tool for long-running operations.
- Verify that your VCF Business Services console role has sufficient permissions. See Access Management.

The following procedure uses the VCF Download Tool to download VMware Cloud Foundation 9.1.1 installation, upgrade binaries, artifacts, ESX binaries, and metadata to software depot in offline depot mode and involves the following steps.

Download the latest version of the VCF Download Tool on a computer with access to the Internet or the offline depot web server.

- Generate an activation code from the VCF Business Services console.
- Download installation, upgrade, and ESX binaries and metadata.
- If the VCF Download Tool and web server are on separate machines, transfer the downloaded binaries to the offline depot.

Note:

For artifacts that contains OCI images, you must upload the artifacts to software depot by using the VCF Download Tool. See Upload OCI Images When Software Depot Is in Offline Mode.

Important Considerations

- The VCF Download Tool is the only supported method for downloading binaries for VCF 9.1 and later in environments without Internet access. Do not use third-party tools or other methods to download binaries.
- If you have a vSphere Foundation license, in the following commands, replace `--sku VCF` with `--sku VVF`.
- If the computer hosting the offline depot can only access the Internet using a proxy server, you can use proxy settings when downloading binaries. See Proxy Server Options.
- VMware Cloud Foundation does not support proxy servers with Kerberos or NTLMv2 authentication.

In the steps below, where applicable, replace the following values.

Value	Description
vcf-source-version	Your current VCF version. For example, `9.1.1.0`.
`/path/to/binaries`	The download location for the binaries. For example, `/var/www`.
`/path/to/activation-code.txt`	Path to a text file that contains a single line with your activation code from the VCF Business Services Console.

1. Download VCF Download Tool 9.1.1 or later to a machine with Internet access or the offline depot web server.
   a) Log in to the Broadcom Support Portal and browse to **My Downloads** > **VMware Cloud Foundation**.
   b) Click the version of VMware Cloud Foundation to which you are upgrading.
   c) Click **Drivers & Tools**.
   d) Click the download icon for the VCF Download Tool.
   e) Extract `vcf-download-tool-<version.build>.tar.gz` to the offline depot web server.
   f) Navigate to the `vcf-download-tool-<version.build>/bin/` and confirm that you have execute permission on all folders.

      The directory includes both the Bundle Transfer Utility (`lcm-bundle-transfer-util`) and the VCF Download Tool (`vcf-download-tool`).
2. Obtain and enter a software depot activation code for VMware Cloud Foundation Download Tool.
   a) Generate and copy a software depot ID.

      ```
      vcf-download-tool configuration generate --software-depot-id
      ```
   b) Log in to the VCF Business Services console.
   c) In the upper-left corner of the console, next to the tenant name, click the arrow to expand the section.
   d) From the drop-down menu, select the correct tenant name and siteID that map to your VCF or vSphere Foundation entitlement.
   e) Navigate to **Software Depot Registrations** > **New Registration** and use your software depot ID (`software-depot-id`) to obtain an activation code (`depotDownloadActivationCode`) for your depot.
   f) Enter the activation code that you obtained from the VCF Business Services console into a file, for example, `/path/to/activation-code.txt`.
3. List the available installation and upgrade binaries.
   - List installation binaries.

     ```
     ./vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install
     ```
   - List upgrade binaries.

     ```
     ./vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type UPGRADE
     ```

In VCF Download Tool 9.1.1 and later, by default, you list only the latest available versions of the binaries. To list all available versions, you can add the `--latest false` filter option.

- In VCF Download Tool 9.1.1 and later, list artifacts.

  ```
  ./vcf-download-tool artifacts list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt
  ```

4. (Optional) Configure the VCF Download Tool to download patches and notifications from a third-party vendor.

   In the following command, replace `https://<host_URL>/index.xml` with the third-party vendor URL address.

   ```
   ./bin/vcf-download-tool esx configuration --add-url https://<host_URL>/index.xml
   ```

5. Download binaries and metadata to the machine with Internet access or machine hosting the offline depot.

   - Download all installation binaries.

     ```
     ./vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install --depot-store /path/to/binaries
     ```

   - Download specific installation binaries. For example, VCF Automation.

     ```
     ./vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --component VRA --type INSTALL --depot-store /path/to/binaries
     ```

     For a detailed list of components, see **Filter Options** table in VCF Download Tool Binaries Commands and Options.

   - Download upgrade binaries.

     ```
     ./vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type UPGRADE --depot-store /path/to/binaries
     ```

   In VCF Download Tool 9.1.1 and later, by default, you download only the latest available versions of the binaries. To download all available versions, you can add the `--latest false` filter option.

   - Download ESX binaries and metadata.

     ```
     ./vcf-download-tool esx download --depot-download-activation-code-file /path/to/activation-code.txt --depot-store /path/to/binaries
     ```

     Note:

     Downloading ESX binaries and metadata might take some time to complete and a significant number of files can be downloaded during this process. You can monitor the progress by reviewing the log file at `./logs/esximage.log`.

   - In VCF Download Tool 9.1.1 and later, download all artifacts.

     ```
     ./vcf-download-tool artifacts download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --depot-store /path/to/binaries
     ```

   - In VCF Download Tool 9.1.1 and later, download specific artifacts. For example, vSphere Supervisor.

     ```
     ./vcf-download-tool artifacts download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --component SUPERVISOR --depot-store /path/to/binaries
     ```

     For a detailed list of components, see **Filter Options for Artifacts** table in VCF Download Tool Artifacts Commands.

6. If VCF Download Tool runs on a different machine than the web server, transfer the downloaded binaries to the machine hosting the offline depot.

- Connect your software depot to the offline depot. See Configure a Software Depot Connection Mode.
- After the binaries and ESX components are available, use the VCF Operations user interface to apply the binaries to workload domains and use the vSphere Client to create a vSphere Lifecycle Manager image for upgrading ESX. See Download Binaries to Software Depot.

Download Binaries to Software Depot in Disconnected Mode by Using the VCF Download Tool

If software depot does not have access to an online or offline depot, you can use the VCF Download Tool to download binaries and ESX component data to a different computer and then upload the binaries and data to software depot.

- A Windows or Linux computer with Internet connectivity (either directly or through a proxy) for downloading the binaries and other required files.
- Configure TCP keepalive in your SSH client to prevent socket connection timeouts for long-running operations.
- Verify that your VCF Business Services console role has sufficient permissions. See Access Management.

The following procedure uses the VCF Download Tool to transfer VMware Cloud Foundation 9.1.1 installation, upgrade binaries, artifacts, ESX binaries, and metadata to software depot in disconnected mode, and involves the following steps.

- Download the latest version of the VCF Download Tool on a computer with access to the Internet.
- Generate an activation code from the VCF Business Services console.
- Download installation, upgrade, and ESX binaries and metadata, to the computer with Internet access.
- From a machine with access to the Internet, upload the VMware Cloud Foundation binaries and ESX binaries to software depot.

Important Considerations

- The VCF Download Tool is the only supported method for downloading binaries for VCF 9.1 and later in environments without Internet access. Do not use third-party tools or other methods to download binaries.
- If you have a vSphere Foundation license, in the following commands, replace `--sku VCF` with `--sku VVF`.
- If the computer with Internet access can only access the Internet using a proxy server, you can use proxy settings when downloading binaries. See Proxy Server Options.
- VMware Cloud Foundation does not support proxy servers with Kerberos or NTLMv2 authentication.

In the steps below, where applicable, replace the following values.

Value	Description
vcf-source-version	Your current VCF version. For example, `9.1.1.0`.
`/path/to/binaries`	The download location for the binaries. For example, `/var/www`.
/path/to/upload_file.tar	The path to the tar file, containing the exported VCF binaries.
`/path/to/activation-code.txt`	Path to a text file that contains a single line with your activation code from the VCF Business Services Console.
OPS-FQDN	Enter the fully qualified domain name (FQDN) of the primary VCF Operations node.
FLEET-FQDN	Enter the fully qualified domain name (FQDN) of the fleet-level components.
username	Enter your VCF Operations username, for example, `admin`.
ops-password.txt	Enter the path to the VCF Operations password file.

1. Download VCF Download Tool 9.1.1 or later on a computer with Internet access.
   a) Log in to the Broadcom Support Portal and browse to **My Downloads** > **VMware Cloud Foundation**.
   b) Click the version of VMware Cloud Foundation to which you are upgrading.
   c) Click **Drivers & Tools**.
   d) Click the download icon for the VCF Download Tool.
   e) Extract `vcf-download-tool-<version.build>.tar.gz`.
   f) Navigate to the `vcf-download-tool-<version.build>/bin/` and confirm that you have execute permission on all folders.

      The directory includes both the Bundle Transfer Utility (`lcm-bundle-transfer-util`) and the VCF Download Tool (`vcf-download-tool`).
2. Obtain and enter a software depot activation code for VMware Cloud Foundation Download Tool.
   a) Generate and copy a software depot ID.

      ```
      vcf-download-tool configuration generate --software-depot-id
      ```
   b) Log in to the VCF Business Services console.
   c) In the upper-left corner of the console, next to the tenant name, click the arrow to expand the section.
   d) From the drop-down menu, select the correct tenant name and siteID that map to your VCF or vSphere Foundation entitlement.
   e) Navigate to **Software Depot Registrations** > **New Registration** and use your software depot ID (`software-depot-id`) to obtain an activation code (`depotDownloadActivationCode`) for your depot.
   f) Enter the activation code that you obtained from the VCF Business Services console into a file, for example, `/path/to/activation-code.txt`.
3. List the available installation and upgrade binaries for a target version.
   - List installation binaries.

     ```
     ./vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install
     ```
   - List upgrade binaries.

     ```
     ./vcf-download-tool binaries list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type UPGRADE
     ```

   In VCF Download Tool 9.1.1 and later, by default, you list only the latest available versions of the binaries. To list all available versions, you can add the `--latest false` filter option.

   - In VCF Download Tool 9.1.1 and later, list artifacts.

     ```
     ./vcf-download-tool artifacts list --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt
     ```
4. (Optional) Configure the VCF Download Tool to download patches and notifications from a third-party vendor.

   In the following command, replace `https://<host_URL>/index.xml` with the third-party vendor URL address.

   ```
   ./bin/vcf-download-tool esx configuration --add-url https://<host_URL>/index.xml
   ```
5. Download binaries and metadata to the computer with Internet access.
   - Download all installation binaries for the target version.

     ```
     ./vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type INSTALL --automated-install --depot-store /path/to/binaries
     ```
   - Download specific installation binaries for the target version. For example, VCF Automation.

     ```
     ./bin/vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --component VRA --type INSTALL --depot-store /path/to/binaries
     ```

For a detailed list of components, see **Filter Options** table in VCF Download Tool Binaries Commands and Options.

- Download upgrade binaries.

```
./vcf-download-tool binaries download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --type UPGRADE --depot-store /path/to/binaries
```

In VCF Download Tool 9.1.1 and later, by default, you download only the latest available versions of the binaries. To download all available versions, you can add the `--latest false` filter option.

- Download ESX binaries and metadata.

```
./vcf-download-tool esx download --depot-download-activation-code-file /path/to/activation-code.txt --depot-store /path/to/binaries
```

Note:

Downloading ESX binaries and metadata might take some time to complete and a significant number of files can be downloaded during this process. You can monitor the progress by reviewing the log file at `./logs/esximage.log`.

- In VCF Download Tool 9.1.1 and later, download all artifacts.

```
./vcf-download-tool artifacts download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --depot-store /path/to/binaries
```

- In VCF Download Tool 9.1.1 and later, download specific artifacts. For example, vSphere Supervisor.

```
./vcf-download-tool artifacts download --sku VCF --vcf-version vcf-source-version --depot-download-activation-code-file /path/to/activation-code.txt --component SUPERVISOR --depot-store /path/to/binaries
```

For a detailed list of components, see **Filter Options for Artifacts** table in VCF Download Tool Artifacts Commands.

6. From a machine that has access to software depot, upload the binaries.

   The machine with access to software depot can be the same as the computer with access to the Internet. Otherwise, transfer the binaries from the computer with Internet access to the machine with access to software depot.

   - Upload installation, upgrade, and metadata to software depot.

```
./vcf-download-tool depot binaries upload --ops-fqdn OPS-FQDN --ops-auth-source LOCAL --ops-user username --ops-user-password-file ops-password.txt --depot-fqdn FLEET-FQDN --sku VCF --vcf-version vcf-source-version  --depot-store /path/to/binaries
```

   - Upload ESX binaries and metadata to software depot.

```
./vcf-download-tool depot esx upload --ops-fqdn OPS-FQDN --ops-auth-source LOCAL --ops-user username --ops-user-password-file ops-password.txt --depot-fqdn FLEET-FQDN --depot-store /path/to/binaries
```

   - If you are using `tar` files, export the binaries from the binaries download location to a `tar` file and upload the `tar` file to software depot. You must export and upload binaries, ESX, and artifacts data separately.

```
./vcf-download-tool binaries export --sku VCF --vcf-version vcf-source-version  --depot-store /path/to/binaries --depot-file /path/to/upload_file.tar

./vcf-download-tool esx export --depot-store /path/to/binaries --depot-file /path/to/esx_upload_file.tar

./vcf-download-tool artifacts export --sku VCF --vcf-version vcf-source-version --depot-store /path/to/binaries --depot-file /path/to/artifacts_upload_file.tar
```

```
./vcf-download-tool depot binaries upload --ops-fqdn OPS-FQDN --ops-auth-source LOCAL --ops-
user username --ops-user-password-file ops-password.txt --depot-fqdn FLEET-FQDN --depot-file /path/to/
upload_file.tar

./vcf-download-tool depot esx upload --ops-fqdn OPS-FQDN --ops-auth-source LOCAL --ops-user username
 --ops-user-password-file ops-password.txt --depot-fqdn FLEET-FQDN --depot-file /path/to/
esx_upload_file.tar

./vcf-download-tool depot artifacts upload --ops-fqdn OPS-FQDN --ops-auth-source LOCAL --ops-
user username --ops-user-password-file ops-password.txt --depot-fqdn FLEET-FQDN --depot-file /path/to/
artifacts_upload_file.tar
```

- (Optional) List the 9.1.1 binaries that are uploaded to software depot.

```
./vcf-download-tool depot binaries list --ops-fqdn OPS-FQDN --ops-auth-source LOCAL --ops-user username
 --ops-user-password-file ops-password.txt --depot-fqdn FLEET-FQDN --vcf-version vcf-source-version
```

- In VCF Download Tool 9.1.1 and later, upload all artifacts to software depot.

```
./vcf-download-tool depot artifacts upload --ops-fqdn OPS-FQDN --ops-auth-source LOCAL --ops-
user username --ops-user-password-file ops-password.txt --depot-fqdn FLEET-FQDN --sku VCF --vcf-
version vcf-source-version  --depot-store /path/to/binaries
```

Assign a Software Depot to VCF Instances

You can choose which software depot instances are used by the VCF Instances in your environment.

You can assign your first and secondary software depot instances to specific VCF Instances, even when the software depot instance is not deployed on that VCF Instance. For example, if you deployed a secondary software depot instance to a VCF Instance, you can assign that software depot instance to another VCF Instance in the same region. Later, if needed, you can reassign the first software depot instance to any of these VCF instances. For more information about software depot instances, see Binary Management for VMware Cloud Foundation.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Select **Build** > **Lifecycle** > **VCF Management**.
3. On the **Components** tab, from the list of components of your first VCF instance, click **Software depot**.
4. Select **Actions** > **Assign to VCF Instance.**
5. In the **Assign Sofware Depot to VCF Instance** wizard, select the VCF Instances to assign the software depot to.
6. Click **Assign**.

VCF Download Tool Command Reference Information

The VCF download tool is a command-line utility that you use to download and manage VCF component binaries and metadata in air-gapped environments. It provides commands for downloading, uploading, listing, and cleaning up binaries and metadata to help you prepare and maintain VCF artifacts.

The VCF Download Tool provides a suite of commands for downloading, uploading, listing, and cleaning up VCF component and ESX binaries, enabling efficient management of software components for VMware Cloud Foundation releases. With its comprehensive features, the VCF Download Tool helps streamline the process of preparing, managing, and maintaining VCF-related artifacts in your environment.

VCF Download Tool General Options

Use the VCF download tool general options to display help, check the tool version, and manage your Customer Experience Improvement Program (CEIP) participation and proxy server settings.

Help Options

Option	Description
`-h, --help`	Displays all available VCF Download Tool commands and top-level options.
`binaries -h, binaries --help`	Displays the options and filters available for downloading, listing, exporting, and cleaning up VCF component binaries.
`depot -h, depot --help`	Displays the commands available for uploading, listing, and removing artifacts and binaries in a connected software depot.
`metadata -h, metadata --help`	Displays the options available for downloading and managing VCF component metadata.
`releases -h, releases --help`	Displays the options available for listing and filtering VCF release information.
`esx -h, esx --help`	Displays the options available for downloading and managing ESX binaries and depot configurations.
`artifacts -h, artifacts --help`	Starting with VCF Dowload Tool 9.1.1 and later, this option displays the options and filters available for downloading, listing, exporting, and cleaning up workload component artifacts in a local depot store.

Version Option

Option	Description
`-v, --version`	Displays the version of the VCF Download Tool.

Customer Experience Improvement Program (CEIP) Option

The VCF download tool supports the Broadcom Customer Experience Improvement Program (CEIP), which collects product usage data to improve the customer experience. You can activate or deactivate CEIP participation at any time. For more information, see Customer Experience Improvement Program.

Option	Description
`--ceip ENABLE` or `--ceip DISABLE`	Enables or disables participation in CEIP. The first time you run the VCF Download Tool, you are prompted to choose whether to participate. The tool saves your selection as the default for future runs. To change your selection, specify this option again with a different value.

Table 1088: Proxy Server Options

If the machine, where VCF Download Tool is installed, can access the Internet only by using a proxy server, use the following options when using the `binaries`, `metadata`, `releases`, and `esx` tool commands. The `depot` command cannot use the proxy filter. Staring with VCF Download Tool 9.1.1 and later, you can use these options with the `artifacts` command.

Option	Description
`--proxy-server, -s`	Enter the FQDN and port number of the proxy server that the VCF Download Tool uses to access the Internet. Use the format *`FQDN:port`*. The proxy server must be reachable from the machine where the tool is running. For example: `--proxyServer proxy.example.com:3128`.

`--proxy-https`	Specifies that the proxy server uses HTTPS connections. To use this option, add the proxy certificate to the JRE default trust store of the VCF Download Tool. The exact path to the `/cacerts` folder depends on the directory where you extracted the VCF Download Tool `tar` file.
	For example:
	• Unix
	`/opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore <tool root>/jre/lin64/lib/security/cacerts`
	• Windows
	`/opt/obtu/jre/lin64/bin/keytool -importcert -file ca-bundle.crt -keystore <tool root>/jre/win32/lib/security/cacerts`

`--proxy-user, -r`	Enter the username required to authenticate with the proxy server. Use this option when the proxy server requires authentication. For example, `--proxy-user` *`username`*.
`--proxy-user-password-file`	Enter the path to a text file containing the password for the user specified in `--proxy-user`. The file must contain only the password on a single line. For example, `--proxyPasswordFile` *`path/to/password.txt`*.

Table 1089: Filter Options

These filter options can be applied to the following VCF Download Tool commands:

- Downloading Binaries
- Listing Binaries
- Cleaning up Binaries
- Binaries Export
- Depot Binaries Upload
- Depot Binaries List

- Depot Binaries Cleanup

<table>
<tr><th>Option</th><th>Description</th></tr>
<tr><td><code>--automated-install</code></td><td>Filters the operation to binaries required by the VCF Installer.</td></tr>
<tr><td><code>--component component</code></td><td>Filters the operation to binaries for a specified component. Replace <code>component</code> with one of the values in the component values table below.

Table 1090: Component values
<table><tr><th>Component</th><th>Description</th></tr><tr><td>DEPOT_SERVICE</td><td>software depot</td></tr><tr><td>ESX_HOST</td><td>ESX</td></tr><tr><td>HCX</td><td>VCF Operations HCX</td></tr><tr><td>NSX_T_MANAGER</td><td>NSX</td></tr><tr><td>SDDC_MANAGER_VCF</td><td>SDDC Manager</td></tr><tr><td>TELEMETRY_ACCEPTOR</td><td>Telemetry</td></tr><tr><td>VCF_CONSUMPTION_CLI</td><td>VCF Consumption CLI</td></tr><tr><td>VCF_CONSUMPTION_CLI_PLUGINS</td><td>VCF Consumption CLI Plugins</td></tr><tr><td>VCF_FLEET_LCM</td><td>fleet lifecycle</td></tr><tr><td>VCF_LICENSE_SERVER</td><td>license server</td></tr><tr><td>VCF_OBSERVABILITY_DATA_PLATFORM</td><td>real-time metrics</td></tr><tr><td>VCF_OPS_CLOUD_PROXY</td><td>cloud proxy</td></tr><tr><td>VCF_SALT</td><td>Salt server</td></tr><tr><td>VCF_SALT_RAAS</td><td>Salt RaaS</td></tr><tr><td>VCF_SDDC_LCM</td><td>SDDC lifecycle</td></tr><tr><td>VCF_SERVICE_VCD_MIGRATION_BACKEND</td><td>migration service engine</td></tr><tr><td>VCFDT</td><td>VCF download tool</td></tr><tr><td>VCFMS_METRICS_STORE</td><td>real-time metrics store</td></tr><tr><td>VCENTER</td><td>vCenter</td></tr><tr><td>VIDB</td><td>identity broker</td></tr><tr><td>VMRC</td><td>VMware Remote Console</td></tr><tr><td>VMTOOLS</td><td>VMware Tools</td></tr><tr><td>VRA</td><td>VCF Automation</td></tr><tr><td>VRLI</td><td>log management</td></tr><tr><td>VRNI</td><td>VCF Operations for networks</td></tr><tr><td>VRO</td><td>VCF Operations orchestrator</td></tr><tr><td>VROPS</td><td>VCF Operations</td></tr><tr><td>VRSLCM</td><td>Operations fleet management (VCF 9.0 only)</td></tr><tr><td>VSP</td><td>vSphere Supervisor Platform</td></tr><tr><td>VSAN_ESA_WITNESS</td><td>vSAN ESA Witness</td></tr><tr><td>VSAN_FILE_SERVICES</td><td>vSAN File Services</td></tr><tr><td>VSAN_OSA_WITNESS</td><td>vSAN OSA Witness</td></tr></table>For more information, see Components in VCF and vSphere Foundation.</td></tr>
</table>

Option	Description
`--component-version ComponentVersion`	Filters the operation to binaries for a specific component version. Replace `ComponentVersion` with the version number to filter by.
`--lifecycle-managed-by lcm`	Filters the operation to binaries whose lifecycle is managed by the specified component. Replace `lcm` with one of the following values: • SDDC_MANAGER_VCF (SDDC Manager) • VRSLCM (VCF Operations fleet management, 9.0 only) • VCF_FLEET_LCM (Fleet Lifecycle) • SELF (Self-managed)
`--patches-only`	Filters the operation to binaries for patch versions of components only.
`--sku sku`	Filters the operation to binaries for the specified product SKU. Replace `sku` with `VCF` for VMware Cloud Foundation or `VVF` for VMware vSphere Foundation.
`-t type` or `--type type`	Filters the operation to binaries of the specified type. Replace `type` with `INSTALL` for installation binaries or `UPGRADE` for upgrade binaries.
`--upgrades-only`	Filters the operation to binaries for upgrade versions of components only.
`--vcf-version version[..[end version]]`	Filters the operation to binaries for a specific VCF version or version range. You can use the following formats: Downloads / Lists / Cleans up / Exports binaries for a specific version `version`of VMware Cloud Foundation, or for a range of versions . You can enter the version string in the following formats: • `a.b`: For example, if you enter 9.0, the tool downloads the binaries for 9.0.0.0 and any 9.0.x binaries. • `a.b.c`: For example, if you enter 9.0.1, the tool downloads all 9.0.1.x binaries. • `a.b.c.d`: For example, if you enter 9.0.1.0, the tool only downloads 9.0.1.0 binaries. • For a version range, specify `start_version..end_version`. The tool includes binaries for both the start and end versions.
`--id bundleIds`	Filters the operation to one or more specific binaries by bundle ID. Replace `bundleIds` with a single bundle ID or a comma-separated list of bundle IDs.
`--download-spec-file downloadSpecFile`	Filters the download to only the binaries required for a specific planned upgrade. Replace `downloadSpecFile` with the path to the download specification file generated by the `create-download-spec` command.
`--latest`	In VCF Download Tool 9.1.1 and later, filters the operation to only the latest available versions of the binaries (such as the latest Express Patches for required components). This behavior is enabled by default. To override this and include all available versions, specify the `--latest false` filter option.

VCF Download Tool Binaries Commands and Options

Use the VCF Download Tool `binaries` commands to download, upload, list, clean up, and export VCF installation, upgrade, and patch binaries.

Binaries Commands

Command	Description
`download`	Downloads VCF binaries and metadata, including the unified release manifest, product version catalog, vSAN HCL, and compatibility data. At least one filter option is required. When you run the command multiple times against the same output directory, the tool adds newly downloaded binaries to the existing content without overwriting it.
`upload`	Uploads downloaded binaries and metadata to software depot.
`list`	Lists all available VCF binaries. At least one filter option is required. For each binary, the command displays the component name, version number, release date, size, and type. Use this command to identify relevant binaries before downloading.
`cleanup`	Removes unwanted or obsolete binaries from the local depot store while retaining binaries you still need. Run this command on a machine where you have already downloaded binaries.
`create-download-spec`	Generates a download specification file based on the planned upgrades for a specific workload domain. Use the generated file as input to the `download` command with `--download-spec-file` to download only the binaries required for the planned upgrade.
`export`	Archives locally downloaded binaries and metadata into a tar file for transfer to another system.

With VCF 9.1.1 and later, to reduce storage requirements, for example, when there are multiple express patches for a VCF component version, only the latest patch for each component is downloaded. To download all the patches, you can add the `--latest` *`false`* filter option. This option default to true and is applicable to the `download`, `upload`, `list`, and `export` commands.

Binaries Download Options

You can run several download commands with the same *`depotStore`* and the result collects the binaries.

Table 1091: Depot Directory Options for Downloading Binaries

Option	Description
`-d` *`depotStore`* or `--depot-store` *`depotStore`*	Replace *`depotStore`* with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.

Table 1092: Depot Authentication Options for Downloading Binaries

Option	Description
`--depot-download-token-file` *`depotDownloadTokenFile`*	Replace *`depotDownloadTokenFile`* with the path to text file containing single line with depot activation code. Activation code can be generated at the Broadcom download portal and used for authentication.
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

Table 1093: Filter Options for Downloading Binaries

For filter options, see Filter Options.
For proxy settings, see Proxy Server Options.

Binaries Upload Options

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded binaries and metadata.
`-f sddcManagerFqdn` or `--sddc-manager-fqdn sddcManagerFqdn`	Replace `sddcManagerFqdn` with the FQDN of the SDDC Manager instance. The VCF download tool uses this address to connect to SDDC Manager.
`-u sddcManagerUsername` or `--sddc-manager-user sddcManagerUsername`	Replace `sddcManagerUsername` with the username for connecting to SDDC Manager.
`--sddc-manager-user-password-file sddcManagerPasswordFile`	Replace `sddcManagerPasswordFile` path to a text file containing the password for the user specified in `--sddc-manager-user`. The file must contain only the password on a single line. If you do not provide this file, the tool prompts you to enter the password interactively.

Binaries List Options

Table 1094: Depot Authentication Options for Listing Binaries

Option	Description
`--depot-download-token-file depotDownloadTokenFile`	Replace `depotDownloadTokenFile` with the path to text file containing single line with depot activation code. Activation code can be generated at the Broadcom download portal and used for authentication.
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

Table 1095: Filter Options for Listing Binaries

For filter options, see Filter Options.
For proxy settings, see Proxy Server Options.

Binaries Cleanup Options

Table 1096: Depot Options for Cleaning up Binaries

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded binaries and metadata.

Table 1097: Filter Options for Cleaning up Binaries

For filter options, see Filter Options.

Binaries Download Specification File Options

Option	Description
`--domain-name` *domainNames*	Replace *domainNames* with a workload domain name or a comma-separated list of workload domain names for which to generate the download specification. If you do not specify a domain name, the tool fetches all available domains.
`--download-spec-dir` *downloadSpecFile*	Replace *downloadSpecFile* with the path to the directory where the tool saves the download specification file generated by the `create-download-spec` command.
`-f` *sddcManagerFqdn* or `--sddc-manager-fqdn` *sddcManagerFqdn*	Replace *sddcManagerFqdn* with the FQDN of the SDDC Manager instance. The VCF download tool uses this address to connect to SDDC Manager.
`-u` *sddcManagerUsername* or `--sddc-manager-user` *sddcManagerUsername*	Replace *sddcManagerUsername* with the username for connecting to SDDC Manager.
`--sddc-manager-user-password-file` *sddcManagerPasswordFile*	Replace *sddcManagerPasswordFile* path to a text file containing the password for the user specified in `--sddc-manager-user`. The file must contain only the password on a single line. If you do not provide this file, the tool prompts you to enter the password interactively.

Binaries Export Options

Option	Description
`-d` *depotStore* or `--depot-store` *depotStore*	Replace *depotStore* with the path to the directory where the VCF download tool stores downloaded binaries and metadata.
`--depot-file` *depotFile*	Replace *depotFile* with the path to a tar archive containing the depot contents to upload.

Table 1098: Filter Options for Binaries Export

For filter options, see Filter Options.

VCF Download Tool Software Depot Management Options

Use the VCF download tool `depot` commands to upload binaries and metadata to software depot, list available content, and clean up obsolete binaries from software depot.

Depot Commands

Command	Description
`binaries`	Manages binary files in software depot. Supports upload, cleanup, and list subcommands.
`metadata`	Manages metadata files in software depot. Supports upload subcommands.
`esx`	Manages ESX binary and metadata files in software depot. Supports upload subcommands.
`artifacts`	Manages workload component metadata and artifacts binaries in software depot.

Depot Binaries Options

Command	Description
`upload`	Uploads metadata and binaries to software depot. Provide either a tar archive of binaries using `--depot-file`, or a depot store path with `--depot-store` together with filter options to select which binaries to upload.
`cleanup`	Removes unwanted or obsolete binaries from software depot while retaining the binaries you still need. Run this command on a machine that already has binaries downloaded to the depot store.
`list`	Lists all binaries available in software depot. At least one filter option is required. For each binary, the command displays the component name, version number, release date, size, and type. Use this command to assess software depot contents before downloading or uploading.

Table 1099: Depot Binaries Authentication Options for `upload`, `cleanup`, and `list` Commands

Option	Description
`--ops-fqdn` *opsFqdn*	Replace *opsFqdn* with the FQDN of the VCF Operations instance. The VCF download tool uses this address to authenticate and obtain access tokens for software depot API calls.
`--ops-auth-source` *opsAuthSource*	Replace *opsAuthSource* with the authentication source for the VCF Operations username and password. Use `LOCAL` for a VCF Operations local account, or enter the name of an identity provider configured in VCF Operations. Defaults to `LOCAL` if not specified.
`--ops-user` *opsUser*	Replace *opsUser* with the username for authenticating with VCF Operations or a VCF Operations integrated identity provider. This user must have the permissions required to access software depot resources.
`--ops-user-password-file` *opsUserPasswordFile*	Replace *opsUserPasswordFile* the path to a text file containing the password for the user specified in `--ops-user`. The file must contain only the password on a single line.
`--api-token-endpoint-url` *apiTokenEndpointUrl*	Replace *apiTokenEndpointUrl* with the API token endpoint URL for fetching the access token used to authenticate software depot API calls.
`--api-token-file` *apiTokenFile*	Replace *apiTokenFile* with the path to a text file containing the API token for authentication. The file must contain only the token on a single line.

Table 1100: Depot Binaries Upload Options

Option	Description
`-d` *depotStore* or `--depot-store` *depotStore*	Replace *depotStore* with the path to the directory where the VCF download tool stores downloaded binaries and metadata.
`--depot-file` *depotFile*	Replace *depotFile* with the path to a tar archive containing the depot contents to upload.
`-f` or `--force`	Force restart upload from the beginning, discarding any existing progress.
`--depot-fqdn` *depotFqdn*	Replace *depotFqdn* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1101: Depot Binaries Upload Filter Options

For filter options, see Filter Options.

Table 1102: Depot Binaries Cleanup Options

Option	Description
`-f` or `--force`	Force restart cleanup from the beginning, discarding any existing progress.
`--depot-fqdn` *`depotFqdn`*	Replace *`depotFqdn`* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1103: Depot Binaries Cleanup Filter Options

For filter options, see Filter Options.

Table 1104: Depot Binaries List Options

Option	Description
`--depot-fqdn` *`depotFqdn`*	Replace *`depotFqdn`* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1105: Depot Binaries List Filter Options

For filter options, see Filter Options.

Depot Metadata Options

Command	Description
`upload`	Upload the metadata and binaries to software depot. Either provide a depot file containing archived binaries, or provide a depot store path where binaries and metadata are present and bundle filters to create and upload the necessary binaries.

Table 1106: Depot Metadata Upload Options

Option	Description
`-d` *`depotStore`* or `--depot-store` *`depotStore`*	Replace *`depotStore`* with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.
`--depot-file` *`depotFile`*	Replace *`depotFile`* with the path to a tar archive containing the depot contents to upload.
`-f` or `--force`	Force restart upload from the beginning, discarding any existing progress.
`--depot-fqdn` *`depotFqdn`*	Replace *`depotFqdn`* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1107: Depot Metadata Upload Authentication Options

Option	Description
`--ops-fqdn` *opsFqdn*	Replace *opsFqdn* with the FQDN of the VCF Operations instance. The VCF download tool uses this address to authenticate and obtain access tokens for software depot API calls.
`--ops-auth-source` *opsAuthSource*	Replace *opsAuthSource* with the authentication source for the VCF Operations username and password. Use `LOCAL` for a VCF Operations local account, or enter the name of an identity provider configured in VCF Operations. Defaults to `LOCAL` if not specified.
`--ops-user` *opsUser*	Replace *opsUser* with the username for authenticating with VCF Operations or a VCF Operations integrated identity provider. This user must have the permissions required to access software depot resources.
`--ops-user-password-file` *opsUserPasswordFile*	Replace *opsUserPasswordFile* the path to a text file containing the password for the user specified in `--ops-user`. The file must contain only the password on a single line.
`--api-token-endpoint-url` *apiTokenEndpointUrl*	Replace *apiTokenEndpointUrl* with the API token endpoint URL for fetching the access token used to authenticate software depot API calls.
`--api-token-file` *apiTokenFile*	Replace *apiTokenFile* with the path to a text file containing the API token for authentication. The file must contain only the token on a single line.

Depot ESX Options

Command	Description
`upload`	Uploads ESX metadata and binaries to software depot. Provide either a tar archive of binaries using `--depot-file`, or a depot store path with `--depot-store` together with filter options to select which ESX binaries to upload.

Table 1108: Depot ESX Upload Options

Option	Description
`-d` *depotStore* or `--depot-store` *depotStore*	Replace *depotStore* with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.
`--depot-file` *depotFile*	Replace *depotFile* with the path to a tar archive containing the depot contents to upload.
`-f` or `--force`	Force restart upload from the beginning, discarding any existing progress.
`--depot-fqdn` *depotFqdn*	Replace *depotFqdn* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1109: Depot ESX Upload Authentication Options

Option	Description
`--ops-fqdn` *opsFqdn*	Replace *opsFqdn* with the FQDN of the VCF Operations instance. The VCF download tool uses this address to authenticate and obtain access tokens for software depot API calls.
`--ops-auth-source` *opsAuthSource*	Replace *opsAuthSource* with the authentication source for the VCF Operations username and password. Use `LOCAL` for a VCF Operations local account, or enter the name of an identity provider configured in VCF Operations. Defaults to `LOCAL` if not specified.
`--ops-user` *opsUser*	Replace *opsUser* with the username for authenticating with VCF Operations or a VCF Operations integrated identity provider. This user must have the permissions required to access software depot resources.
`--ops-user-password-file` *opsUserPasswordFile*	Replace *opsUserPasswordFile* the path to a text file containing the password for the user specified in `--ops-user`. The file must contain only the password on a single line.
`--api-token-endpoint-url` *apiTokenEndpointUrl*	Replace *apiTokenEndpointUrl* with the API token endpoint URL for fetching the access token used to authenticate software depot API calls.
`--api-token-file` *apiTokenFile*	Replace *apiTokenFile* with the path to a text file containing the API token for authentication. The file must contain only the token on a single line.

Depot Artifacts Options

Command	Description
`upload`	Uploads metadata and artifacts to software depot. Provide either a tar archive of binaries using `--depot-file`, or a depot store path with `--depot-store` together with filter options to select which binaries to upload.
`list`	Lists all metadata and artifacts available in software depot. At least one filter option is required. For each binary, the command displays the component name, version number, release date, size, and type. Use this command to assess software depot contents before downloading or uploading.
`cleanup`	Removes unwanted or obsolete metadata and artifacts from software depot. Run this command on a machine that already has metadata and binaries downloaded to the depot store.

Table 1110: Depot Artifacts Upload Options

Option	Description
`-d` *depotStore* or `--depot-store` *depotStore*	Replace *depotStore* with the path to the directory where the VCF download tool stores downloaded binaries and metadata.
`--depot-file` *depotFile*	Replace *depotFile* with the path to a tar archive containing the depot contents to upload.
`-f` or `--force`	Force restart upload from the beginning, discarding any existing progress.
`--depot-fqdn` *depotFqdn*	Replace *depotFqdn* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1111: Depot Artifacts Authentication Options for `upload`, `list`, and `cleanup` Commands

Option	Description
`--ops-fqdn` *opsFqdn*	Replace *opsFqdn* with the FQDN of the VCF Operations instance. The VCF download tool uses this address to authenticate and obtain access tokens for software depot API calls.
`--ops-auth-source` *opsAuthSource*	Replace *opsAuthSource* with the authentication source for the VCF Operations username and password. Use `LOCAL` for a VCF Operations local account, or enter the name of an identity provider configured in VCF Operations. Defaults to `LOCAL` if not specified.
`--ops-user` *opsUser*	Replace *opsUser* with the username for authenticating with VCF Operations or a VCF Operations integrated identity provider. This user must have the permissions required to access software depot resources.
`--ops-user-password-file` *opsUserPasswordFile*	Replace *opsUserPasswordFile* the path to a text file containing the password for the user specified in `--ops-user`. The file must contain only the password on a single line.
`--api-token-endpoint-url` *apiTokenEndpointUrl*	Replace *apiTokenEndpointUrl* with the API token endpoint URL for fetching the access token used to authenticate software depot API calls.
`--api-token-file` *apiTokenFile*	Replace *apiTokenFile* with the path to a text file containing the API token for authentication. The file must contain only the token on a single line.

Table 1112: Depot Aritifacts Upload Filter Options

For filter options, see Filter Options for Artifacts.

Table 1113: Depot Artifacts List Options

Option	Description
`--depot-fqdn` *depotFqdn*	Replace *depotFqdn* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1114: Depot Artifacts List Filter Options

For filter options, see Filter Options for Artifacts.

Table 1115: Depot Artifacts Cleanup Options

Option	Description
`-f` or `--force`	Force restart cleanup from the beginning, discarding any existing progress.
`--force-delete-shared`	Force delete all selected shared artifacts without asking for consent.
`--depot-fqdn` *depotFqdn*	Replace *depotFqdn* with the FQDN of the software depot instance. The VCF download tool uses this address to connect to software depot.

Table 1116: Depot Artifacts Cleanup Filter Options

For filter options, see Filter Options for Artifacts.

VCF Download Tool Metadata Commands and Options

Use the VCF download tool `metadata` commands to download and upload VCF metadata files, including the unified release manifest, product version catalog, vSAN HCL, and compatibility data.

Metadata Commands

Command	Description
`download`	Downloads all metadata required by software depot. in a single operation, including the unified release manifest, product version catalog, vSAN HCL, and compatibility data. The tool automatically retrieves all relevant metadata files. **Note:** This command does not download ESX metadata. Use the `esx metadata` command for ESX metadata.
`upload`	Uploads all downloaded metadata to software depot. Before running this command, transfer the downloaded metadata to a machine that can reach software depot.
`export`	Archives locally downloaded metadata into a tar file for transfer to another system.

Metadata Download Options

If the computer with Internet access can only access the Internet using a proxy server, use the following options when downloading metadata. For proxy settings, see Proxy Server Options.

Table 1117: Depot Directory Options for Downloading Metadata

Option	Description
`-d` *`depotStore`* or `--depot-store` *`depotStore`*	Replace *`depotStore`* with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.

Table 1118: Depot Authentication Options for Downloading Metadata

Option	Description
`--depot-download-token-file` *`depotDownloadTokenFile`*	Replace *`depotDownloadTokenFile`* with the path to text file containing single line with depot activation code. Activation code can be generated at the Broadcom download portal and used for authentication.
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

Metadata Upload Options

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded binaries and metadata.
`-f sddcManagerFqdn` or `--sddc-manager-fqdn sddcManagerFqdn`	Replace `sddcManagerFqdn` with the FQDN of the SDDC Manager instance. The VCF download tool uses this address to connect to SDDC Manager.
`-u sddcManagerUsername` or `--sddc-manager-user sddcManagerUsername`	Replace `sddcManagerUsername` with the username for connecting to SDDC Manager.
`--sddc-manager-user-password-file sddcManagerPasswordFile`	Replace `sddcManagerPasswordFile` path to a text file containing the password for the user specified in `--sddc-manager-user`. The file must contain only the password on a single line. If you do not provide this file, the tool prompts you to enter the password interactively.

Metadata Export Options

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded binaries and metadata.
`--depot-file depotFilePath` or `-d depotFilePath`	Replace `depotFilePath` the path and filename for the tar archive that the tool creates during export.

VCF Download Tool Releases Commands

Use the VCF download tool `releases` commands to view a list of available VCF releases and the product details for each release.

Releases Commands

Command	Description
`list`	Lists all available VCF releases. When you run the command without specifying a version, it returns all available releases. When you specify a version, it returns the products included in that release. The command supports filtering with `--vcf-version`.

Releases Options

If the computer with Internet access can only access the Internet using a proxy server, use the following options when listing releases:

Table 1119: Depot Authentication Options for Release Commands

Option	Description
`--depot-download-token-file depotDownloadTokenFile`	Replace `depotDownloadTokenFile` with the path to text file containing single line with depot activation code. Generate the activation code on the Broadcom download portal and use it to authenticate requests.
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

Table 1120: Filter Options for Release Commands

Option	Description
`--vcf-version` *`version`*	Filters the results to a specific VCF version. Replace *`version`* with the version number to display the products included in that release. If you do not specify a version, the command returns all available VCF versions.
For proxy settings, see Proxy Server Options.	

VCF Download Tool ESX Commands

Use the VCF download tool `esx` commands to download and manage ESX binaries and metadata. Use the downloaded data to create vSphere Lifecycle Manager images in the vSphere Client.

ESX Commands

The `esx` commands replace the functionality previously provided by Update Manager Download Service (UMDS).

Command	Description
`download`	Downloads ESX binaries and metadata from the configured depot sources.
`export`	Exports downloaded ESX binaries and metadata into a tar file for transfer to another system.
`import`	Imports existing ESX offline bundles or patch store archives into the local depot store.
`metadata`	Downloads ESX metadata, including the VVS bundle, from the configured depot sources.
`configuration`	Manages ESX download configuration, including depot source URLs and enabled host platforms.

ESX Download Options

Table 1121: Depot Directory Options for Downloading ESX Binaries

Option	Description
`-d` *`depotStore`* or `--depot-store` *`depotStore`*	Replace *`depotStore`* with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.
`-f` or `--force`	Cleans up existing content in the depot store before downloading from the configured sources.
For proxy settings, see Proxy Server Options.	

Table 1122: Depot Authentication Options for Downloading ESX Binaries

Option	Description
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

ESX Export Options

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded ESX binaries and metadata.
`--depot-file depotFile`	Replace `depotFile` with the path to a tar archive containing the depot contents to upload.

ESX Import Options

You can use this option to import any existing ESX bundles you have.

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the absolute path to the offline zip bundle file to import.

Table 1123: ESX Import Options for Depot Configuration

Option	Description
`-F offlineBundle` or `--offline-bundle offlineBundle`	Replace `offlineBundle` with the absolute path to the offline zip bundle file for import operation.
`-S importStore` or `--import-store importStore`	Replace `importStore` with the absolute path to the XML file of the source patch store to import.

ESX Metadata Options

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.
`-f` or `--force`	Option to clean up existing content and then download from the specified sources.
For proxy settings, see Proxy Server Options.	

Table 1124: Depot Authentication Options for Downloading ESX Metadata

Option	Description
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

ESX Configuration Options

Option	Description
`--add-url depotUrlToAdd`	Replace `depotUrlToAdd` with the URL of a custom or third-party online depot source to add to the ESX download configuration.
`-D hostPlatformsToDisable` or `--disable-host hostPlatformsToDisable`	Replace `hostPlatformsToDisable` with a comma-separated list of host platform identifiers to disable. Disabled platforms are excluded from `download` and `export` operations.

Option	Description
`-E` or `--enable-host` *`hostPlatformsToEnable`*	Replace *`hostPlatformsToDisable`* with a comma-separated list of host platform identifiers to enable. Enabled platforms are included in `download` and `export` operations.
`-G` or `--get`	Displays the current ESX download configuration, including configured depot source URLs and enabled host platforms.
`--remove-url` *`depotUrlToRemove`*	Replace *`depotUrlToRemove`* with the URL of a custom or third-party online depot source to remove from the ESX download configuration.

VCF Download Tool Artifacts Commands

Starting with VCF Dowload Tool 9.1.1 and later, you can use the `artifacts` commands to list, download, export, and clean up workload component artifact binaries in a local depot store on a machine with access to the Internet.

Artifacts Commands

Artifacts are the binary files, OCI images, and associated metadata that is required to install, upgrade, and manage VCF services such as Supervisor, VKS, and VCF Consumption CLI. Unlike the `binaries` commands, which download VCF core and management component binaries, the `artifacts` commands target workload component artifacts - the software packages that run on top of the VCF platform.

You use the `artifacts` commands to download, list, clean up, and export these files to a local depot store for use in air-gapped environments.

Command	Description
`download`	Downloads all workload component artifacts and their associated metadata. At least one filter option is required. The command supports filtering to narrow the selection of artifacts based on the intended operation. When you run the command multiple times against the same output directory, the tool adds newly downloaded artifacts to the existing content without overwriting it.
`list`	Lists all available workload component artifacts. At least one filter option is required. For each artifact, the command displays the component name, version number, size, and type. Use this command to identify relevant artifacts before downloading.
`cleanup`	Removes unwanted or obsolete workload component artifacts from the local depot store while retaining the artifacts you still need. Run this command on a machine where you have already downloaded artifacts.
`export`	Exports locally downloaded workload component artifacts into a `tar` file for transfer to another system.

Artifacts Download Options

Table 1125: Depot Directory Options for Downloading Artifacts

Option	Description
`-d` *`depotStore`* or `--depot-store` *`depotStore`*	Replace *`depotStore`* with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.
For proxy settings, see Proxy Server Options.	

Table 1126: Depot Authentication Options for Downloading Artifacts

Option	Description
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

For proxy settings, see Proxy Server Options.

Table 1127: Filter Options for Artifacts

<table>
<tr><th>Option</th><th>Description</th></tr>
<tr><td><code>--category categoryType</code></td><td>The category type for which to download artifacts bundles. Replace <code>categoryType</code> with SUPERVISOR, VKS, SUPERVISOR_SERVICE, VCF_CLI, or VCF_SERVICE.</td></tr>
<tr><td><code>--component component</code></td><td>Component type for which to download artifacts. Replace <code>component</code> with one of the following values.
Some of the components contain OCI images, which are downloaded as TAR files.
<table>
<tr><th>Component value</th><th>Artifact name</th></tr>
<tr><td>SUPERVISOR</td><td>Supervisor</td></tr>
<tr><td>SUPERVISOR_SERVICE_VKS</td><td>vSphere Kubernetes Service</td></tr>
<tr><td>SUPERVISOR_SERVICE_LCI</td><td>Local Consumption Interface</td></tr>
<tr><td>SUPERVISOR_SERVICE_ARGOCD</td><td>Argo CD Supervisor Service</td></tr>
<tr><td>SUPERVISOR_SERVICE_CONTOUR</td><td>Contour Supervisor Service</td></tr>
<tr><td>SUPERVISOR_SERVICE_SUPERVISOR_MANAGEMENT_PROXY</td><td>Supervisor Management Proxy</td></tr>
<tr><td>SUPERVISOR_SERVICE_METRICS_AGGREGATOR</td><td>Metrics Aggregator</td></tr>
<tr><td>SUPERVISOR_SERVICE_CA_CLUSTERISSUER</td><td>CA Cluster Issuer</td></tr>
<tr><td>SUPERVISOR_SERVICE_EXTDNS</td><td>External DNS Supervisor Service</td></tr>
<tr><td>SUPERVISOR_SERVICE_HARBOR</td><td>Harbor (Supervisor)</td></tr>
<tr><td>SUPERVISOR_SERVICE_NATIVE_OBJECT_STORE</td><td>Native Object Store</td></tr>
<tr><td>VKR</td><td>vSphere Kubernetes Release</td></tr>
<tr><td>VKS_STANDARD_PACKAGES</td><td>VKS Standard Packages</td></tr>
<tr><td>VKSM_EXTENSIONS</td><td>VKSM Extensions</td></tr>
<tr><td>VCF_SERVICE_ARGOCD</td><td>Argo CD</td></tr>
<tr><td>VCF_SERVICE_HARBOR</td><td>Harbor</td></tr>
<tr><td>VCF_SERVICE_SECRET_STORE</td><td>Secret Store Service</td></tr>
<tr><td>VCF_SERVICE_METRICS_AGGREGATOR</td><td>Metrics Aggregator</td></tr>
<tr><td>VCF_SERVICE_DATA_SERVICES</td><td>Data Services</td></tr>
<tr><td>VCF_SERVICE_PROTECTION_AND_RECOVERY</td><td>Protection and Recovery</td></tr>
<tr><td>VCF_SERVICE_VKSM_AUTO_ATTACH</td><td>VKSM Auto Attach</td></tr>
<tr><td>VCF_SERVICE_CONFIGURATION</td><td>Configuration Service</td></tr>
<tr><td>VCF_SERVICE_ENCRYPTION_MANAGEMENT</td><td>Encryption Management</td></tr>
<tr><td>VCF_SERVICE_VSAN_BACKEND</td><td>vSAN Backend</td></tr>
<tr><td>VCF_SERVICE_NATIVE_OBJECT_STORE</td><td>Native Object Store</td></tr>
<tr><td>VCF_CONSUMPTION_CLI</td><td>VCF Consumption CLI</td></tr>
</table>
</td></tr>
</table>

Option	Description
`--component-version ComponentVersion`	Filters the operation to binaries for a specific component version. Replace `ComponentVersion` with the version number to filter by.
`--sku sku`	Filters the operation to binaries for the specified product SKU. Replace `sku` with `VCF` for VMware Cloud Foundation or `VVF` for VMware vSphere Foundation.
`--vcf-version version[..[end version]]`	Filters the operation to binaries for a specific VCF version or version range. You can use the following formats: Downloads / Lists / Cleans up/ Exports binaries for a specific version `version`of VMware Cloud Foundation, or for a range of versions . You can enter the version string in the following formats: • `a.b`: For example, if you enter 9.0, the tool downloads the binaries for 9.0.0.0 and any 9.0.x binaries. • `a.b.c`: For example, if you enter 9.0.1, the tool downloads all 9.0.1.x binaries. • `a.b.c.d`: For example, if you enter 9.0.1.0, the tool only downloads 9.0.1.0 binaries. • For a version range, specify `start_version..end_version`. The tool includes binaries for both the start and end versions.
`--id bundleIds`	Filters the operation to one or more specific binaries by bundle ID. Replace `bundleIds` with a single bundle ID or a comma-separated list of bundle IDs.

Artifacts List Options

Table 1128: Depot Authentication Options for Listing Artifacts

Option	Description
`--depot-download-activation-code-file /path/to/activation-code.txt`	Replace `/path/to/activation-code.txt` with the path to a text file containing your activation code on a single line. Generate an activation code in the VCF Business Services console.

Table 1129: Filter Options for Listing Artifacts

For filter options, see *Filter Options for Downloading Artifacts*.
For proxy settings, see Proxy Server Options.

Artifacts Cleanup Options

Table 1130: Depot Options for Cleaning up Artifacts

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.
`--force-delete-shared`	Force delete shared artifacts without warning.

Table 1131: Filter Options for Cleaning up Artifacts

For filter options, see *Filter Options for Downloading Artifacts*.

Artifacts Export Options

Option	Description
`-d depotStore` or `--depot-store depotStore`	Replace `depotStore` with the path to the directory where the VCF download tool stores downloaded binaries and metadata. This directory is the local repository for all downloaded artifacts and must maintain the required directory structure.
`--depot-file depotFile`	Replace `depotFile` with the depot contents archived in tar format.

Table 1132: Filter Options for Artifacts Export

For filter options, see *Filter Options for Downloading Artifacts*.

Lifecycle Management of VCF Components

Use VCF Operations to upgrade or patch VCF components.

Before you can deploy, upgrade, or patch the VCF management components, you must download the necessary binaries to software depot See Binary Management for VMware Cloud Foundation.

After downloading the necessary binaries, you can proceed with deployment, upgrade, or patch.

- For deployment, see: Completing the Deployment of Your Platform and Adding or Removing VCF Components Post Deployment.
- For upgrading to a major or a minor version, see: Upgrading to VMware Cloud Foundation 9.1.x.
- For applying maintenance or patch releases, see Upgrade or Patch an Individual VCF Component.

You can also use software depot to manage ESX components and vSphere Lifecycle Manager images.

Forward-Only Upgrade Requirement

VCF enforces strict upgrade path validation. You can only upgrade your VCF components to a target version that is released after the release date of your current version. The target version can be a major, minor, or a maintenance release. In general, you can upgrade directly to a patch release only if you can also upgrade to its parent maintenance release. For example:

- Supported: VCF 5.2.0 (25 JUN 2024) to VCF 9.1.0 (11 MAY 2026).
- Not supported: VCF 5.2.4 (27 MAY 2026) to VCF 9.1.0 (11 MAY 2026).
- Supported: VCF 5.2.4 (27 MAY 2026) to VCF 9.1.1 (03 SEP 2026).

This is valid for all components in a VCF source BoM. For example:

- Supported upgrade to VCF 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3a (18 JUN 2024) and ESX 8.0 Update 3 (25 JUN 2024), which are part of VCF 5.2.0 BoM.
- Not supported upgrade to VCF 9.1.0 (11 MAY 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026), which are part of VCF 5.2.4 BoM.
- Supported upgrade to VCF 9.1.1 (03 SEP 2026): vCenter 8.0 Update 3j (27 MAY 2026) and ESX 8.0 Update 3j (27 MAY 2026), which are part of VCF 5.2.4 BoM.

VCF components might have further limitations, see VMware Interoperability Matrix.

Upgrade and Patch Order for VCF Components

Operation Type	Version Change	Upgrade or Patch Order
Upgrade	A change in the major (9.0 to 10.0) or minor (9.0 to 9.1) version of a VCF component. See VCF Lifecycle Operation Types.	Major and minor releases require a strict upgrade order: • For management domain upgrade order, see Upgrading to VMware Cloud Foundation 9.1.x. • For workload domain upgrade order, see Upgrading VMware Cloud Foundation Workload Domains to 9.1.x.
Patch	A change in the maintenance (9.1.0 to 9.1.1) or patch (9.0.1.0200 to 9.0.1.0300) version of a VCF component. See VCF Lifecycle Operation Types.	Maintenance and patch releases do not require a strict patch order, unless the release notes explicitly instruct you to follow a specific order. The 9.1.1 maintenance release has specific component interdependencies. See Patching VCF from 9.1.0.x to 9.1.1.

Maintenance and patch releases do not affect component interoperability. You apply maintenance and patch releases as per the needs of your organization, unless the release notes explicitly instruct you to follow a specific order.

Before you apply a maintenance or a patch release, plan for possible downtime of certain functionalities performed by the component and consider the impact on other services in the SDDC. For example, when patching software depot, no other lifecycle operations can be performed, due to their dependency on the availability of install, upgrade, and patch binaries.

For more information on the functionalities of the components, see Components in VCF and vSphere Foundation.

Patching VCF from 9.1.0.x to 9.1.1

VCF 9.1.1 has specific patching dependencies between VCF components:

1. You begin applying the 9.1.1 maintenance release in your 9.1.0.x environment by patching the VCF management services fleet lifecycle component to 9.1.1 before any other VCF component.
2. After the fleet lifecycle component is at version 9.1.1, you must comply only with the following dependencies:
   a. VCF Operations must not be patched in parallel with other components. Before you patch VCF Operations, ensure that no other lifecycle operations in the environment are in progress. Do not start other lifecycle operations until the VCF Operations patch process completes.
   b. Before you patch ESX hosts from 9.1.0.x to 9.1.1.0, you must first patch the VCF Operations instance and the license servers connected to it.
   c. Before you patch an identity broker instance from 9.1.0.x to 9.1.1, you must first patch the respective VCF management services runtime that hosts the identity broker instance.
   d. Before you patch Salt RaaS from 9.1.0.x to 9.1.1, you must first patch the respective VCF management services runtime that hosts the Salt RaaS instance.
   e. Before you patch the migration service engine component from 9.1.0.x to 9.1.1, you must first patch VCF Automation.
   f. When you patch a software depot instance, patching other components is blocked because the patch binaries are unavailable until the patch process for the software depot completes.

License Server and Cloud Proxy

The lifecycle management of the license server and the cloud proxy components is bundled with the lifecycle management of VCF Operations. Upgrading or patching VCF Operations automatically upgrades or patches license server and cloud proxy.

Shared NSX Instances

For NSX instances that are shared across multiple domains, you must upgrade the NSX, vCenter, and ESX components in all domains that share the NSX instance. If you cancel the automatically created upgrade plan for a dependent domain,

the upgrade plan for all domains that share the NSX instance is also canceled. If the upgrade plan is canceled, you must create another upgrade plan.

For example, if you create a domain upgrade plan or shared NSX upgrade plan for domain A, you must add and upgrade the vCenter instance and all ESX hosts in this domain. Before you can finalize the upgrade, you must visit domain B, add the vCenter and ESX components to the component upgrade plan, and upgrade all three components. After you upgrade the vCenter, ESX, and NSX components in domain B and all other dependent domains, you can finalize the upgrade in domain A and the dependent domains.

> **Note:**
>
> If an NSX 9.1 instance is shared between a management domain and workload domain(s), when all other components are upgraded, you must start the **VMware NSX Manager Nodes Upgrade** and **VMware NSX Edge Cluster Upgrade and NSX Upgrade Finalize** upgrade steps from the management domain only.

VCF Component Version Tracking

To verify the current versions of your VCF components, in VCF Operations, you click **Build** > **Lifecycle** > **VCF Instance**, then select a VCF domain. The **Component Versions** tab displays information for supported components, such as current version, target version, and status for each component. The tab displays version information for cluster hosts and standalone hosts, and compares installed versions with target versions to identify version drift. For clusters with ESX hosts that run different versions, a warning appears. The displayed cluster version represents the minimum version across all hosts in the cluster.

Tracked VCF Components

- SDDC Manager
- vCenter
- NSX Manager
- ESX clusters and hosts
- Standalone ESX hosts
- HCX

Option	Description
On Target	The current component version matches the target version.
Version Drift	The current component version differs from the target version.

Target versions are defined when you create a domain or component upgrade plan. When no target version is configured, the current version is used as the target, and the status displays as **On Target**.

Upgrade or Patch an Individual VCF Component

Use VCF Operations to upgrade or apply a patch to an individual VCF component.

- If you are upgrading or patching VCF core components you must download the respective patch binaries. The method that you use to download binaries depends on how you access the Internet in your environment. See Binary Management for VMware Cloud Foundation. If you are connected to the online depot, binaries are available in the patch planner. If you do not have access to an online depot, use the VCF Download Tool to download binaries. See Download Binaries to an Offline Depot by Using the VCF Download Tool or Download Binaries to Software Depot in Disconnected Mode by Using the VCF Download Tool.
- Before you upgrade or patch, back up the component. See Component Backup and Restore of VMware Cloud Foundation.

You can upgrade a single component to a maintenance release. You can apply a patch release to a single component at a time. If an upgrade or a patch operation is in progress, you cannot upgrade or patch another component.

1. Log in to VCF Operations.
2. On the top navigation bar, click **Build**.
3. In the left navigation pane, click **Lifecycle**.
4. To upgrade or patch a VCF management or core component, follow the respective procedure.

VCF Management Components	1. In the **Lifecycle** pane, click **VCF Management** and click the **Upgrade** tab. 2. Click **Change target version**. 3. In the **Set Target Version** dialog box, from the version drop-down menu, select the patch version for the corresponding major version and click **Customize**. 4. For the VCF component to patch, select a version from the **Target version** drop-down menu, and click **Set version**. 5. In the **Management component target version** section, for the VCF component to patch, click **Run prechecks**. 6. After the prechecks complete, remediate any errors. 7. In the **Management component target version** section, for the VCF component to patch, click **Upgrade**.
VCF Core Components	1. Expand **VCF Instances** and select a domain in your VCF instance. 2. Click the **Upgrades** tab and click the **Plan Component Upgrade** button. You cannot use the patch planner if you have an existing upgrade plan. Cancel the upgrade plan to create a patching plan. 3. The **Plan Component Upgrade** wizard opens. 4. On the **Select Components and Target Version**, select the required component and its target version, and click **Next**. 5. On the **Plan Overview** page, click **Submit Plan**. 6. In the **Upgrade Sequence** section, in the component card, click **Run Prechecks**. 7. Resolve any issues before proceeding with an upgrade. 8. In the **Upgrade Sequence** section, in the component card, to upgrade the selected component, click **Schedule** or **Start Now**.

Upgrade or Patch SDDC Manager from VCF Operations

You can upgrade or patch only SDDC Manager without having to upgrade the entire VCF Instance. An independent SDDC Manager patch release includes a change in the fourth group of digits in its version number, for example, `SDDC Manager 9.0.0.0100`.

- Download the SDDC Manager upgrade or patch binary from software depot. See Download Binaries to Software Depot.

You can upgrade SDDC Manager without upgrading the VCF Instance when:

- The target version of SDDC Manager is compatible with all the BOM versions running in your VCF Instance (management and workload domains).
- There is a supported upgrade path from your current SDDC Manager version to the target SDDC Manager version.

> **Note:**
> You can use the SDDC Manager upgrade functionality to upgrade SDDC Manager even when the target version of SDDC Manager is part of a full VCF BOM release, as long as it is compatible.

Updating only SDDC Manager, does not change the version of the management domain.

1. Log in to VCF Operations.
2. Select **Build** > **Lifecycle**.
3. Expand **VCF Instances** and the available VCF Instances, select the management domain that hosts the SDDC Manager instance that you want to patch or upgrade.
4. Click the **Upgrades** tab and click **Plan component upgrade**.

   The **Plan component upgrade** wizard opens.

5. On the **Select components and target version** page, select **SDDC Manager**, select a target version, and click **Next**.
6. On the **Plan overview** page, click **Submit plan**.
7. To verify the readiness of your environment agains the selected target version, click **Run prechecks** .
8. To start the patch or upgrade now or schedule for a later time, click **Upgrade now** or **Schedule**.

Managing vSphere Lifecycle Manager Images for VMware Cloud Foundation

VMware Cloud Foundation uses vSphere Lifecycle Manager images to apply software and firmware updates to the ESX hosts in an SDDC cluster.

You create vSphere Lifecycle Manager images in the vSphere Client and then import them into a VCF Instance using VCF Operations. You can use imported images when you:

- Upgrade the management domain or a workload domain.
- Create a workload domain.
- Add an SDDC cluster to the management domain or a workload domain.

Clusters can have up to four additional image definitions to make on a single composite image of the cluster.

The base ESX version is static for all additional images and cannot be customized. The vendor addon, firmware, and components of each image definition can be customized to accommodate clusters with mixed hardware. Each additional image definition can have a unique hardware support manager (HSM) attached.

You can create a composite image for a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. A composite image contains one default image and one or more alternative images. All images in a composite image contain the same base image and solutions, but may include different vendor or firmware and driver add-ons.

Additional images can be manually assigned to a subset of hosts in a cluster, or automatically using System BIOS information including Vendor, Model, OEM String and Family values.

For more information about vSphere Lifecycle Manager images, see Managing Host and Cluster Lifecycle in the vSphere documentation.

Import a vSphere Lifecycle Manager Image to a VCF Instance

After you create vSphere Lifecycle Manager images in the vSphere Client, you can import those images into VCF Operations and use them for creating and updating VCF domains and clusters.

Create a vSphere Lifecycle Manager image using the vSphere Client. See How to Use the Image Library in a vCenter Instance.

- Use the ESX version specified in the VMware Cloud Foundation BOM.
- You can create a composite image for a cluster that has hosts from different vendors or from the same vendor but different generations, family, or model. A composite image contains one default image and one or more alternative images. All images in a composite image contain the same base image and solutions, but may include different vendor or firmware and driver add-ons. When you create alternate images, you also specify the rules that are used to filter the hosts to which the alternative image is assigned.
- If you plan to use the **Import from a file** option to import a vSphere Lifecycle Manager image, you must first export the files from the vSphere Client. See How to Use the Image Library in a vCenter Instance.

1. Log in to VCF Operations at `https://<vcf_operations_fqdn>` with a user assigned the **Administrator** role.
2. Navigate to **Build** > **Lifecycle** and expand **VCF Instances**.
3. Select your VCF Instance.
4. Click **Image Management** and select **Import Image**.
5. Select the import method, provide the required information, and click **Import**.

Option	Description
Import from a vCenter	Select the vCenter where you created the image from the drop-down menu. Select the image to import.
Import from a file	Enter a name for the image and then click **Select File** to import each of the required files. Select the JSON, ZIP, and ISO files from your local computer. The ISO file is optional.

Upgrading VMware Cloud Foundation Workload Domains to 9.1.x

To upgrade to a VMware Cloud Foundation workload domain to 9.1.x, all workload domains must be at VMware Cloud Foundation 5.2 or later. If any workload domain is at a version lower than 5.2, you must upgrade it to 5.2 and then upgrade to 9.1.x.

You can use VCF Operations or the SDDC Manager UI to upgrade your VMware Cloud Foundation 5.2 workload domains to 9.1.x. The following procedures use the SDDC Manager UI.

> **Note:**
> The SDDC Manager UI is being deprecated and will be removed in a future release. After upgrading your workload domains to 9.0, use VCF Operations to perform lifecycle management of VMware Cloud Foundation components.

The steps and the order of steps to upgrade a workload domain are similar to the steps to upgrade the management domain.

Task	Procedure
1. Plan the upgrade.	Plan a Domain Upgrade
2. Perform an upgrade precheck.	Perform Upgrade Prechecks
3. Upgrade NSX in a federated environment.	Upgrading NSX Global Manager Nodes in a Federated Environment
4. Upgrade NSX and vCenter.	Upgrading vCenter and NSX Manager
6. Transition from vLCM baselines to vLCM images.	Transitioning from vSphere Lifecycle Manager Baselines to vSphere Lifecycle Manager Images
7. Upgrade vSphere Supervisor on clusters that have Workload Management enabled.	Updating vSphere Supervisor
8. Upgrade ESX.	Upgrade ESX to 9.1
9. Upgrade vSphere Distributed Switch version	Upgrade vSphere Distributed Switch Versions
10. Upgrade vSAN on-disk format versions.	Upgrade vSAN On-Disk Format Versions

Procedure

1. Log in to SDDC Manager.
2. In the navigation pane, click **Inventory** > **Workload Domains**.
3. On the **Workload Domains** page, click the workload domain you want to upgrade and click the **Upgrades** tab.
4. Under **Available Upgrades**, click **PLAN UPGRADE**.
5. On the **Plan Upgrade for VMware Cloud Foundation** screen, select the target version from the drop-down, and click **NEXT**.

After Completing the Task

After all upgrades have completed successfully:

1. Remove the VM snapshots you took before starting the upgrade.
2. Take a backup of the newly installed components.